ZCAS UNIVERSITY

BACHELOR OF INFORMATION TECHNOLOGY/BACHELOR OF NETWORK ENGINEERING

CIT1142 EMERGING TRENDS IN ICT

END OF SEMESTER FINAL EXAMINATION

TUESDAY 28TH NOVEMBER 2023

12:30 – 15:30 HOURS

Time allowed: 3 HOURS plus 5 Minutes reading time.

Instructions to Candidates:

1. **Read the instructions very carefully.**
2. **Check that you have the correct examination paper in front of you.**
3. **There are FIVE (5) questions in this paper. Section A is Compulsory, and you are required to answer ONLY THREE (3) questions in Section B.**
4. **All questions must be answered in the answer booklet provided only.**
5. **Begin answering each question on a new page.**
6. **No books, files, mechanical/electronic aids, or scientific calculators are permitted in the examination room.**
7. **There shall be NO communication among students during the examination. Any student caught doing this will be disqualified.**

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

SECTION A : Answer the following question.

QUESTION 1.

Read the following then answer the questions that follow.

Scenario:

Alex, an investigator specializing in computer forensics, has been assigned to a case involving a suspected data breach at a small business. The company's confidential information, including customer data and financial records, may have been compromised. Alex begins the investigation by examining the company's computer systems.

Required:

a) What steps should Alex take in the initial assessment phase of the computer forensics investigation to ensure the preservation of digital evidence? **(8 Marks)**

b) As Alex identifies potential evidence related to the data breach, what tools and techniques can be employed to properly collect and preserve digital evidence without altering its integrity? **(8 Marks)**

c) Explain why maintaining a strict chain of custody is crucial in computer forensics, and what measures should Alex implement to ensure the integrity of the digital evidence throughout the investigation process, **(8 Marks)**

d) If some of the compromised data appears to be deleted or encrypted, what methods and tools can Alex use for data recovery and analysis to reconstruct the timeline of events leading up to the data breach? **(8 Marks)**

e) During the investigation, Alex discovers evidence that may be admissible in court. What legal considerations and best practices should Alex follow to ensure that the collected digital evidence is admissible and can withstand legal scrutiny? **(8 Marks)**

[Total: 40 Marks]

SECTION B: Answer ONLY three questions from this section.

QUESTION 2.

At a mid-sized company, the IT department has implemented an Intrusion Detection System (IDS) to enhance the organization's cybersecurity posture. The IDS is responsible for monitoring network traffic and identifying potential security incidents. One day, the IDS triggers an alert indicating a possible intrusion attempt. The IT team assigns Sarah, a junior cybersecurity analyst, to investigate the alert and determine whether it's a genuine security threat.

Required:

a) Explain the steps Sarah should take to investigate the IDS alert thoroughly. Outline the key procedures and tools she might use to determine the nature and severity of the potential intrusion.

(4 Marks)

b) Define the concepts of false positives and false negatives in the context of intrusion detection systems. How can Sarah distinguish between a false positive and a genuine security incident during her investigation?

(4 Marks)

c) If Sarah confirms a security incident based on the IDS alert, what should be her immediate response actions? Outline the steps she should take to contain and mitigate the potential threat to the organization's network.

(4 Marks)

d) Discuss the concept of IDS tuning and optimization and provide examples of adjustments Sarah might make to enhance the system's ability to detect and respond to potential intrusions.

(4 Marks)

e) After resolving the security incident, what steps should Sarah take to document her findings and report to the relevant stakeholders, including IT management and potentially law enforcement? Highlight the importance of clear and comprehensive reporting in the context of cybersecurity incidents.

(4 Marks)

[Total: 20 Marks]

QUESTION 3.

You are a computer science specialist tasked with researching and understanding various aspects of computer applications in web services. Your CEO has assigned you a project to explore the fundamentals, technologies, and trends related to web services in the context of computer applications. To complete this project, you will need to conduct interviews with other international experts in the field, conduct research, and analyze the current state of web services in computer applications.

Required:

a) Explain to the CEO how the use of APIs (Application Programming Interfaces) relates to web services, and what role they play in modern web applications. **(4 Marks)**

b) Discuss the concept of REST (Representational State Transfer) and explain its significance in web service development for computer applications. **(4 Marks)**

c) How do SOAP (Simple Object Access Protocol) and RESTful web services differ in terms of communication and data exchange in computer applications? **(4 Marks)**

d) Provide the current trends and emerging technologies in the field of web services and explain how they might impact computer applications in the near future. **(4 Marks)**

e) Explain the significance of microservices architecture in the context of web services, and how it relates to computer applications. **(4 Marks)**

[Total: 20 Marks]

QUESTION 4.

A large e-commerce company, TechTrends, is looking to enhance its business intelligence and customer insights through data mining. The company has collected a vast amount of data over the years, including customer purchase history, website interactions, and feedback. The management team is eager to leverage data mining techniques to uncover valuable patterns, trends, and relationships within this data to make informed business decisions and improve the overall customer experience.

Required:

a) Explain how you would approach the data preparation phase for TechTrends' e-commerce data before applying data mining techniques. **(4 Marks)**

b) List and discuss potential challenges that might arise in cleaning and transforming the diverse data sets collected by TechTrends. **(4 Marks)**

c) Which data mining techniques do you think would be most suitable for analyzing customer purchase history and predicting future buying behavior for TechTrends? **(4 Marks)**

d) How can clustering algorithms be applied to segment TechTrends' customer base for targeted marketing strategies? **(4 Marks)**

e) Provide potential risks or drawbacks that TechTrends should be aware of when relying on data mining for business decisions. **(4 Marks)**

[Total: 20 Marks]

QUESTION 5.

In the rapidly evolving landscape of global finance, banks are facing unprecedented challenges in safeguarding their customers' assets against a rising tide of fraudulent activities. Bank fraud has become a pressing issue, with perpetrators employing increasingly sophisticated methods to exploit vulnerabilities in financial systems. As a junior analyst at a cybersecurity firm specializing in financial fraud detection, you are tasked with examining trends in bank fraud across cities worldwide to enhance preventative measures and protect customers.

Required:

a) What geographical trends do you observe in the prevalence of bank fraud across major cities globally, and are there specific regions showing higher susceptibility to these activities? **(4 Marks)**

b) How have bank fraud techniques evolved in recent years, and what emerging methods are being employed by perpetrators to exploit vulnerabilities in different cities? **(4 Marks)**

c) Are there discernible patterns in terms of demographics being targeted by bank fraudsters in various cities, and how can this information be utilized to tailor preventive measures? **(4 Marks)**

d) How can international collaboration and information sharing between financial institutions and law enforcement agencies be improved to mitigate the cross-border nature of modern bank fraud? **(4 Marks)**

e) In what ways can advancements in technology be harnessed to enhance the security measures employed by banks, and how can financial institutions strike a balance between innovation and safeguarding customer assets against fraud? **(4 Marks)**

[Total: 20 Marks]

END OF SEMESTER FINAL EXAMINATION